



EXAM PAPERS PRACTICE

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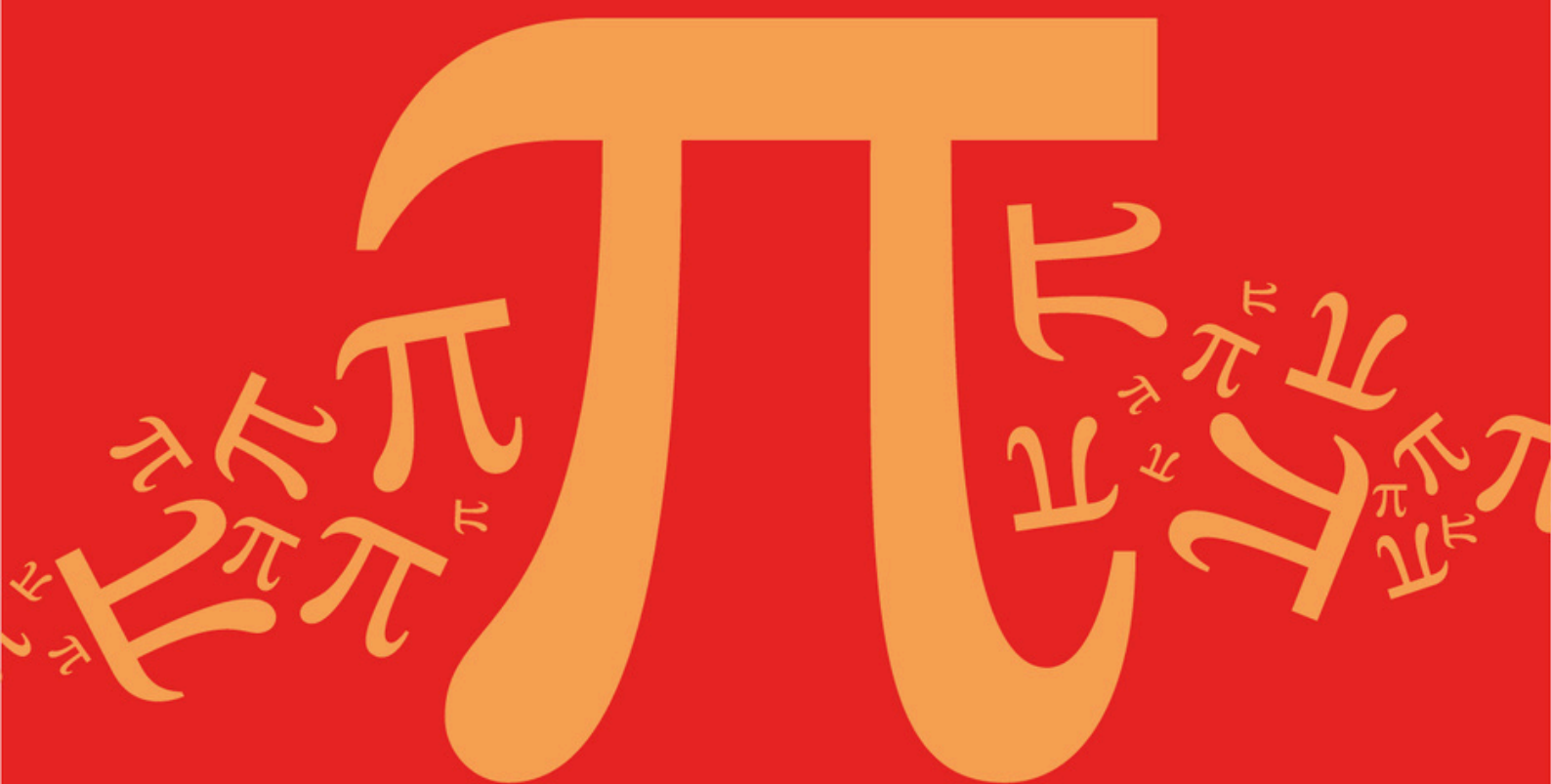
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AQA A Level Further Mathematics (7367)



Topic

B: Complex Numbers

Sub topic

B3: Roots Occur in Conjugate Pairs

Topic Questions

AQA A Level Further Mathematics (7367)

Question Paper

Topic

B: Complex Numbers

Sub topic

B3: Roots Occur in Conjugate Pairs

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Name : _____

Date : _____

Time : _____

Total Marks Available : _____

Total Marks Received : _____

Q1.

$2 - 3i$ is one root of the equation

$$z^3 + mz + 52 = 0$$

where m is real.

(a) Find the other roots.

(3)

(b) Determine the value of m .

(2)

(Total 5 marks)

Q2.

The equation $x^3 - 8x^2 + cx + d = 0$, where c and d are real numbers, has roots α, β, γ .

When plotted on an Argand diagram, the triangle with vertices at α, β, γ has an area of 8.

Given $\alpha = 2$, find the values of c and d .

Fully justify your solution.

(Total 5 marks)

Q3.

(a) It is given that $z = x + yi$, where x and y are real numbers.

(i) Write down, in terms of x and y , an expression for $(z - 2i)^*$.

(1)

(ii) Solve the equation

$$(z - 2i)^* = 4iz + 3$$

giving your answer in the form $a + bi$.

(5)

(b) It is given that $p + qi$, where p and q are real numbers, is a root of the equation $z^2 + 10iz - 29 = 0$.

Without finding the values of p and q , **state** why $p - qi$ is **not** a root of the equation $z^2 + 10iz - 29 = 0$.

(1)

(Total 7 marks)

Q4.

The cubic equation

$$z^3 + pz + q = 0$$

has roots α , β and γ .

- (a) (i) Write down the value of $\alpha + \beta + \gamma$. (1)

- (ii) Express $\alpha\beta\gamma$ in terms of q . (1)

- (b) Show that

$$\alpha^3 + \beta^3 + \gamma^3 = 3\alpha\beta\gamma$$
 (3)

- (c) Given that $\alpha = 4 + 7i$ and that p and q are real, find the values of :

- (i) β and γ ; (2)

- (ii) p and q . (3)

- (d) Find a cubic equation with integer coefficients which has roots, $\frac{1}{\alpha}$, $\frac{1}{\beta}$ and $\frac{1}{\gamma}$. (3)

(Total 13 marks)

Q5.

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The cubic equation

$$2z^3 + pz^2 + qz + 16 = 0$$

where p and q are real, has roots α , β and γ .

It is given that $\alpha = 2 + 2\sqrt{3}i$.

- (a) (i) Write down another root, β , of the equation. (1)

- (ii) Find the third root, γ . (3)

- (iii) Find the values of p and q . (3)

- (b) (i) Express α in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. (2)

- (ii) Show that

$$(2 + 2\sqrt{3}i)^n = 4^n \left(\cos \frac{2n\pi}{3} + i \sin \frac{2n\pi}{3} \right)$$

(2)

(iii) Show that

$$\alpha^n + \beta^n + \gamma^n = 2^{2n+1} \cos \frac{2n\pi}{3} + \left(-\frac{1}{2} \right)^n$$

where n is an integer.

(3)

(Total 14 marks)

Q6.

The complex number $2 + 3i$ is a root of the quadratic equation

$$x^2 + bx + c = 0$$

where b and c are real numbers.

(a) Write down the other root of this equation.

(1)

(b) Find the values of b and c .

(4)

(Total 5 marks)

Q7.

It is given that α , β and γ satisfy the equations

$$\alpha + \beta + \gamma = 1$$

$$\alpha^2 + \beta^2 + \gamma^2 = -5$$

$$\alpha^3 + \beta^3 + \gamma^3 = -23$$

(a) Show that $\alpha\beta + \beta\gamma + \gamma\alpha = 3$.

(3)

(b) Use the identity

$$(\alpha + \beta + \gamma)(\alpha^2 + \beta^2 + \gamma^2 - \alpha\beta - \beta\gamma - \gamma\alpha) = \alpha^3 + \beta^3 + \gamma^3 - 3\alpha\beta\gamma$$

to find the value of $\alpha\beta\gamma$.

(2)

(c) Write down a cubic equation, with integer coefficients, whose roots are α , β and γ .

(2)

(d) Explain why this cubic equation has two non-real roots.

(2)

- (e) Given that α is real, find the values of α , β and γ .

(4)

(Total 13 marks)

Q8.

The cubic equation

$$z^3 + pz^2 + 25z + q = 0$$

where p and q are real, has a root $\alpha = 2 - 3i$.

- (a) Write down another non-real root, β , of this equation.

(1)

- (b) Find:

- (i) the value of $\alpha\beta$;

(1)

- (ii) the third root, γ , of the equation;

(3)

- (iii) the values of p and q .

(3)

(Total 8 marks)

Q9.

- (a) Solve the following equations, giving each root in the form $a + bi$.

- (i) $x^2 + 16 = 0$;

(2)

- (ii) $x^2 - 2x + 17 = 0$.

(2)

- (b) (i) Expand $(1 + x)^3$

(2)

- (ii) Express $(1 + i)^3$ in the form $a + bi$.

(2)

- (iii) Hence, or otherwise, verify that $x = 1 + i$ satisfies the equation

$$x^3 + 2x - 4i = 0$$

(2)

(Total 10 marks)

Q10.

The cubic equation

$$z^3 + pz^2 + 6z + q = 0$$

has roots α , β , and γ .

- (a) Write down the value of $\alpha\beta + \beta\gamma + \gamma\alpha$. (1)
- (b) Given that p and q are real and that $\alpha^2 + \beta^2 + \gamma^2 = -12$:
- (i) explain why the cubic equation has two non-real roots and one real root; (2)
- (ii) find the value of p . (4)
- (c) One root of the cubic equation is $-1 + 3i$.
Find:
- (i) the other two roots; (3)
- (ii) the value of q . (2)
- (Total 12 marks)

Q11.

- (a) (i) Given that $z^6 - 4z^3 + 8 = 0$, show that $z^3 = 2 \pm 2i$. (2)
- (ii) Hence solve the equation
- $$z^6 - 4z^3 + 8 = 0$$
- giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. (6)
- (b) Show that, for any real values of k and θ ,
- $$(z - ke^{i\theta})(z - ke^{-i\theta}) = z^2 - 2kz\cos\theta + k^2$$
- (2)
- (c) Express $z^6 - 4z^3 + 8$ as the product of three quadratic factors with real coefficients. (3)
- (Total 13 marks)

Q12.

- (a) (i) Calculate $(2 + i\sqrt{5})(\sqrt{5} - i)$. (3)

- (ii) Hence verify that $\sqrt{5} - i$ is a root of the equation

$$(2 + i\sqrt{5})z = 3z^*$$

where z^* is the conjugate of z .

(2)

- (b) The quadratic equation

$$x^2 + px + q = 0$$

in which the coefficients p and q are real, has a complex root $\sqrt{5} - i$.

- (i) Write down the other root of the equation.

(1)

- (ii) Find the sum and product of the two roots of the equation.

(3)

- (iii) Hence state the values of p and q .

(2)

(Total 11 marks)

Q13.

The cubic equation

$$x^3 + px^2 + qx + r = 0$$

where p , q and r are real, has roots α , β and γ .

- (a) Given that

$$\alpha + \beta + \gamma = 4 \quad \text{and} \quad \alpha^2 + \beta^2 + \gamma^2 = 20$$

find the values of p and q .

(5)

- (b) Given further that one root is $3 + i$, find the value of r .

(5)

(Total 10 marks)